



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to Sri Vinayaga Enterprises

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

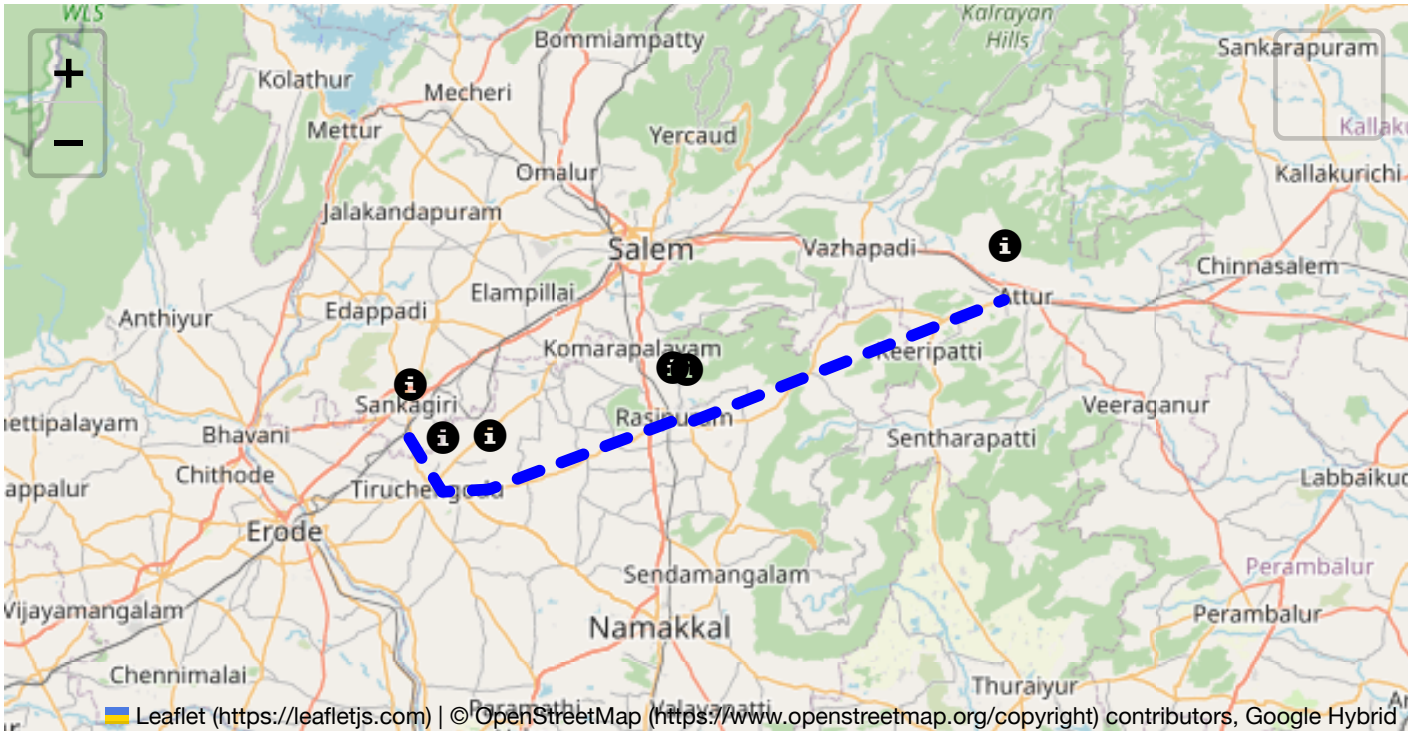
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

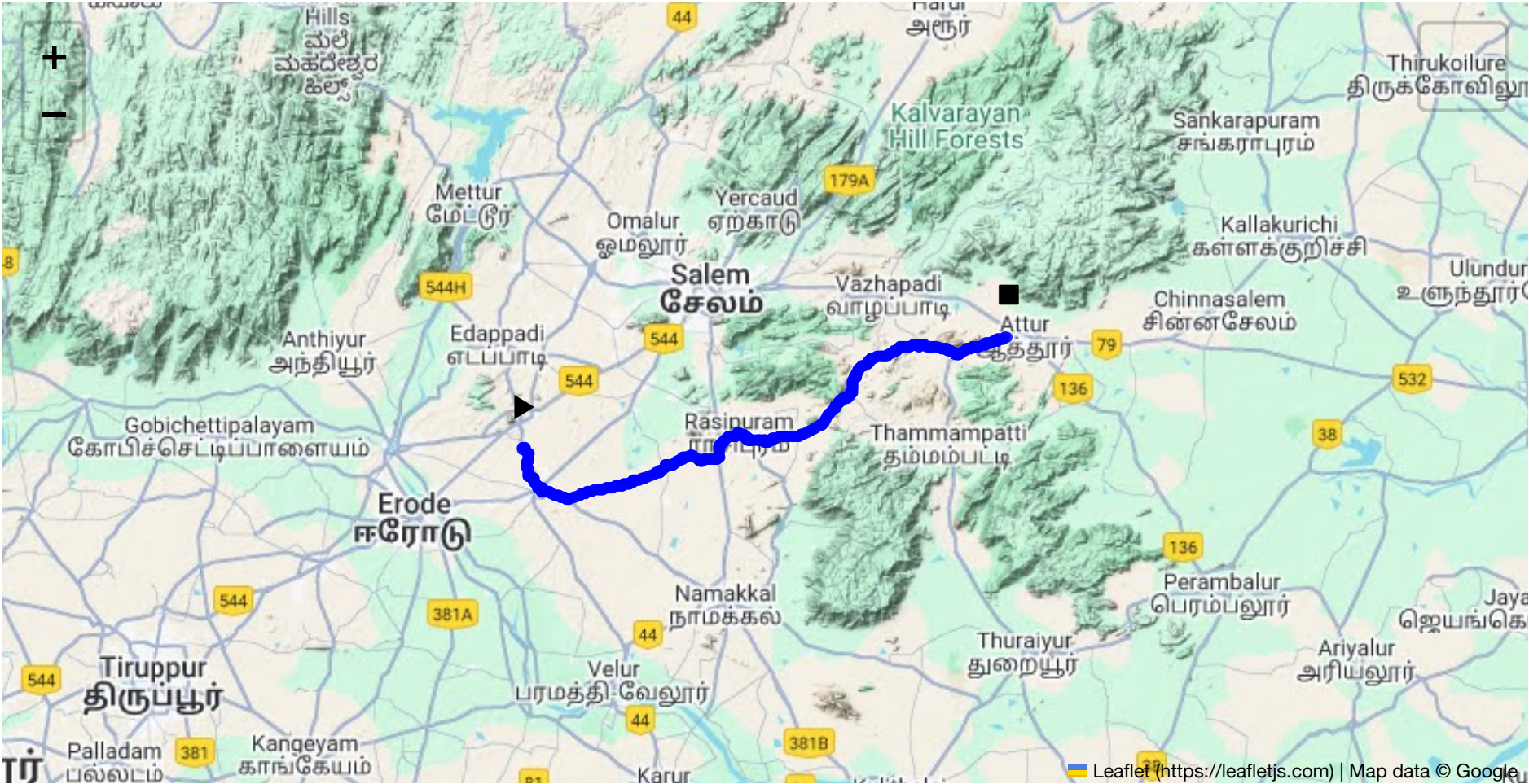
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 97.20 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours

Start: (11.4381, 77.8734)
End: (11.5958249, 78.5702078)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Attur via Namakkal, Kavundampalaiyam, Rasipuram, and Koneripatti covers approximately 97.20 kilometers. The primary roads involved are SH 79 and local roads connecting the various towns and villages. The journey takes around 2.15 hours for heavy vehicles carrying hazardous materials, indicating a route that traverses both state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences tropical weather with hot summers, moderate monsoons, and mild winters. The climate along this route is typically dry but can become hazardous during the monsoon season (June to September) due to heavy rainfall, leading to slippery roads and potential flooding. Summers (April to June) can cause asphalt roads to become quite hot, posing tire safety issues.

3. Analysis of Traffic Patterns

The road from Sangagiri to Namakkal typically sees moderate traffic. Namakkal, being a busy junction, experiences increased congestion, especially during the morning (8:00-10:00 AM) and evening (5:00-7:00 PM) peak hours. Rasipuram and Attur are less congested but can experience local traffic snarls due to markets or local events.

4. Assessment of Road Quality and Infrastructure

The SH 79 is generally well-maintained, but some local roads may have potholes or uneven surfaces, particularly near villages and intersections. Periodic maintenance could cause temporary diversions or single-lane traffic.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or emergencies, alternative routes via the Namakkal-Trichy highway or the Salem-Tiruchirappalli highway can be considered. These routes, although longer, are major connectivity roads and have more reliable infrastructure.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations for the transport of hazardous materials, such as speed limits, restricted hours (usually avoiding night travel in certain areas), and requiring specific permits. Compliance with these rules is crucial to avoid legal and safety issues.

7. Overview of Historical Incidents

While specific historical data on incidents involving heavy vehicles or hazardous materials on this route may not be readily available, Tamil Nadu highways occasionally report accidents due to driver negligence or mechanical failure. Regular maintenance of vehicles and adherence to safety norms significantly reduce the risk.

8. Environmental Considerations and Sensitive Areas

The route passes through both urban and semi-urban areas, with some roads running close to water bodies and agricultural fields. Care should be taken to prevent hazardous material spills, especially near these sensitive zones, to avoid environmental contamination.

9. Analysis of Communication Coverage

The route generally has good mobile network coverage, although certain rural stretches may experience brief dead zones. It is advisable to have alternate communication methods or devices during travel.

10. Estimated Emergency Response Times

Emergency response times can vary. In urban and suburban stretches like Namakkal and Rasipuram, response times may be quick (10-20 minutes). However, in more rural areas between these towns, it might take longer (30 minutes to an hour) due to lesser accessibility.

11. An Overall Summary of Risk Assessment

Overall, the route poses moderate risks primarily due to traffic congestion, road quality issues, and potential weather impacts during the monsoon season. Adhering to local regulations, routine vehicle checks, and being prepared for communication disruptions will mitigate these risks. Drivers should also prepare for longer travel times during peak traffic hours and monsoon periods.

This risk assessment emphasizes the importance of planning, route familiarity, and preparedness in ensuring a safe journey for heavy vehicles transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.4535263, 78.1964113	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	Medium	11.38376, 77.89914	30 KM/Hr
6	Turn	Medium	11.38342, 77.89975	30 KM/Hr
7	Turn	Medium	11.38295, 77.90001	30 KM/Hr
8	Turn	High	11.38186, 77.90258	15 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Turn	Medium	11.36754, 77.93919	30 KM/Hr
11	Turn	Medium	11.41354, 78.08075	30 KM/Hr
12	Turn	High	11.41447, 78.08063	15 KM/Hr
13	Turn	High	11.41475, 78.08087	15 KM/Hr
14	Turn	High	11.42964, 78.11587	15 KM/Hr
15	Turn	Medium	11.42447, 78.15264	30 KM/Hr
16	Turn	Medium	11.42611, 78.15532	30 KM/Hr
17	Turn	Medium	11.42651, 78.15557	30 KM/Hr
18	Turn	High	11.44521, 78.15833	15 KM/Hr
19	Turn	Medium	11.45566, 78.17361	30 KM/Hr
20	Turn	High	11.45587, 78.18057	15 KM/Hr
21	Turn	Medium	11.45760, 78.18140	30 KM/Hr
22	Turn	Medium	11.45990, 78.18413	30 KM/Hr
23	Blind Spot	Blind Spot	11.46035, 78.18534	10 KM/Hr
24	Turn	Medium	11.45929, 78.18544	30 KM/Hr
25	Turn	Medium	11.45895, 78.18572	30 KM/Hr
26	Turn	Medium	11.45856, 78.18775	30 KM/Hr
27	Turn	Medium	11.45824, 78.18811	30 KM/Hr
28	Turn	Medium	11.45764, 78.19208	30 KM/Hr
29	Blind Spot	Blind Spot	11.45100, 78.20385	10 KM/Hr
30	Turn	High	11.45084, 78.20378	15 KM/Hr
31	Blind Spot	Blind Spot	11.45354, 78.19642	10 KM/Hr
32	Turn	Medium	11.45117, 78.20024	30 KM/Hr
33	Turn	Medium	11.44993, 78.22152	30 KM/Hr
34	Turn	Medium	11.44951, 78.22263	30 KM/Hr
35	Turn	High	11.44797, 78.22314	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	Medium	11.44756, 78.22414	30 KM/Hr
37	Turn	Medium	11.45353, 78.23584	30 KM/Hr
38	Turn	Medium	11.49617, 78.32335	30 KM/Hr
39	Turn	Medium	11.49835, 78.32417	30 KM/Hr
40	Turn	Medium	11.50506, 78.33087	30 KM/Hr
41	Turn	Medium	11.50865, 78.33966	30 KM/Hr
42	Turn	Medium	11.51183, 78.34416	30 KM/Hr
43	Turn	Medium	11.51870, 78.34816	30 KM/Hr
44	Turn	Medium	11.54842, 78.35627	30 KM/Hr
45	Turn	Medium	11.56719, 78.37813	30 KM/Hr
46	Turn	Medium	11.56940, 78.39856	30 KM/Hr
47	Turn	Medium	11.57829, 78.51245	30 KM/Hr
48	Turn	Medium	11.58679, 78.54553	30 KM/Hr
49	Turn	Medium	11.59108, 78.55637	30 KM/Hr
50	Turn	Medium	11.59253, 78.55778	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
14	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
15	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
16	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
17	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
18	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
19	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
21	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
22	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
23	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
24	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
26	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
27	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
28	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
29	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
31	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
10	school	SRV EXCEL Matriculation Higher Secondary School	11.4299679, 78.152616	30 km/h	Medium
11	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
13	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
25	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium
30	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.4535263, 78.1964113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38376, 77.89914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38295, 77.90001



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36754, 77.93919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41354, 78.08075



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41447, 78.08063



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41475, 78.08087



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42447, 78.15264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42611, 78.15532



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42651, 78.15557



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44521, 78.15833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45566, 78.17361



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45587, 78.18057



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45760, 78.18140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45990, 78.18413



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46035, 78.18534



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45929, 78.18544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45895, 78.18572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45856, 78.18775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45824, 78.18811



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 78.19208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45100, 78.20385



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45084, 78.20378



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45354, 78.19642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45117, 78.20024



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22152



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44951, 78.22263



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44797, 78.22314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44756, 78.22414



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 78.23584



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49617, 78.32335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49835, 78.32417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50506, 78.33087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50865, 78.33966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51870, 78.34816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54842, 78.35627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56719, 78.37813



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56940, 78.39856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57829, 78.51245



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58679, 78.54553
